

Two-Phone Device Test: 07-10-2026

First end-to-end test of the app on real phones: online recording, offline mesh relay through a guardian, and victim reconnect. It follows the 02-10 and 04-10 build logs. Every result below comes from the app's own logs, captured live from both phones, and from a direct read of the blockchain.

1. Summary

Area	Result	Evidence
Online pipeline: record → encrypt → IPFS → blockchain	PASS	4 clips anchored by the victim phone (21:04–21:08)
Offline relay: victim → guardian over the mesh	PASS	5 of 5 clips and 9 key shares delivered; guardian verified each one
Guardian upload and anchor on the victim's behalf	PASS	5 of 5 uploaded and anchored, using the victim's signature
Automatic retry after a network failure	PASS	Guardian upload failed at 21:30:40 and succeeded on retry at 21:31:45
Victim reconnect: no duplicate anchors	PASS	Chain read at ~21:37: 12 anchor events, every CID exactly once
Reverse relay (guardian → victim)	PASS	Unplanned; worked with no changes (21:36:27–21:37:11)
Two-guardian quorum playback ("Open & verify")	NOT RUN	Needs a third phone
AI distress detection accuracy	ISSUE	Normal speech triggered recordings at least 9 times across both phones

Bugs found: 4 bugs that would have stopped the demo were found and fixed on the day. The worst was a **mesh service-ID mismatch, present since Phase 4**, which meant guardians could never discover victims. Fixes are in commit `d881ce1f`, and all 63 unit/widget tests pass.

2. Setup

Devices		Backend	
Victim	Motorola moto g85 5G (ZA222M8TR7), node <code>aa2f21745858a243</code>	Blockchain	Local Hardhat node (<code>--hostname 0.0.0.0</code>), chain ID 31337
Guardian	Samsung Galaxy A33 (SM-A336E, RZCW10MK16N), node <code>1d928a1910f67f21</code>	Contract	JusticeLedger at <code>0x5FbDB2315678afecb367f032d93F642f64180aa3</code>
Network	Phone hotspot "vivo Y100". PC at <code>10.35.251.51</code> , victim at <code>10.35.251.106</code>	IPFS	Pinata

Build and logging: debug builds installed with `flutter run` over USB. Most of the app logs through `dart:developer`, which neither `flutter run` nor `logcat` shows. A small script was written to stream those records from each phone's Dart VM service, so both phones' upload, anchor, mesh and outbox events could be watched live. Low-level Nearby Connections behaviour was read from `adb logcat`.

3. Work before the device test

- **flutter analyze**: no errors or warnings. 9 info lints: 6 × `avoid_print` in `main.dart`, and 3 × `use_build_context_synchronously` in the guardian scanner.
- **New tests**: `test/shamir_test.dart` (12 tests: 2-of-3 quorum, late-issued shares, tampering, 3-of-5, a hand-computed GF(256) value, random round-trips, validation) and `test/outbox_service_test.dart` (13 tests: dedup, backoff and its 1-hour cap, failure after 20 attempts, `JobNotReady`, job isolation, rerun coalescing, pending mesh jobs). Test-only hooks were added to `OutboxService`.
- **Guardian scanner fix**: after saving, the success snackbar was shown using the closed dialog's context, which throws. The dialog now only returns the name; saving and navigation use the screen's context.
- Committed as `57d0f730`; 62/62 tests passed.

4. Test results

Test 1: Registration PASS

Victim app freshly installed at 20:55:25 and generated Ed25519, X25519 and secp256k1 keys (node `aa2f21745858a243`). The guardian was re-registered at 21:20:40 (node `1d928a1910f67f21`) after its data was wiped (see §6).

Test 2: Pairing both ways PASS

Time	Phone	Event
21:21:53	Victim	Saved guardian <code>1d928a1910f67f21</code>
21:22:06	Guardian	Saved victim <code>aa2f21745858a243</code>

Test 3: Online recording pipeline PASS (after Bug 1 fix)

The first attempts failed: 8 recordings between 20:57:57 and 20:58:52 failed to start with `CameraException(NullPointerException ... ImageReader.getSurface())` (Bug 1). After the fix:

Step	Time	Detail
Recording started	21:02:48	SOS button
Saved	21:03:10	<code>evidence_1791387190804.mp4</code>
Encrypted and sealed	21:03:14	First recording also created the vault key; plaintext deleted
IPFS upload	21:03:49	<code>QmUpiXq51XXtLNK3tCx2pKAFg8UpGzPK28N6W6wcJ1bpDX</code>
Anchored	21:04:07	Tx <code>0xd3496126392424cfe19544d275729c8c59efc665b3877565bba97a31c82bfc24</code>

Three more online clips (all AI-triggered) also completed: `Qmc8LoAR...` (tx `0x434d6748...`, 21:04:12), `QmRBjRvX...` (tx `0x4cc30c51...`, 21:05:10), `QmWAvqqR...` (tx `0x8012c244...`, 21:08:46).

Test 4: Offline mesh relay PASS (after Bug 2 and Bug 3 fixes)

Victim in airplane mode, with Bluetooth and Wi-Fi on but not connected to any network. Uploads failed as expected (No connection to Pinata) and stayed queued. The first attempt found no guardian: advertising had died when Bluetooth toggled (Bug 2), and discovery was scanning the wrong service ID (Bug 3). After both fixes:

Time	Event
21:27:30	Guardian: "Trusted node aa2f21745858a243 nearby. Connecting..."
21:27:35–36	Both: identity verified, secure mesh link established
21:27:36–39	Victim sent key shares for all 9 incidents; guardian confirmed receiving each

21:27:39–44	Victim sent 5 queued clips (file plus signed manifest)
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Clip	Verified and stored by guardian	Uploaded by guardian (CID)	Anchored by guardian (tx)
6b878a39...	21:27:49	21:27:58 QmWvKAXvZyrXCwhA1gMwv4WRrZZWtRJWu3sVXnWSgM HMWW	21:28:02 0xf0e4b8c66 ea5ec52...
6cff3d0d...	21:29:21	21:29:52 QmVaAUqAfeSSgr6c2RmpLPwJzJjhWsMEBpwyELvfKA g6BW	21:29:54 0xe2313d0c8 7359af6...
89248e4a...	21:30:24	Failed 21:30:40 (network); retried 21:31:45 QmRQhfbE7pKF7LJtph 6EuXqJkXDaYgzmeEKrmczp6FWjREV	21:31:47 0xf70b6e70c a9b3de1...
c49afa0e...	21:31:47 (after victim resend)	21:32:25 QmNxvKABYLZiQMt9BjnhXKwfZ15R8YLzzVK3MVfgEE F2YW	21:32:26 0x3fadde497 30fdb20...
e4dd56a6...	21:32:22 (after victim resend)	21:32:45 QmVTAqFUG83Ak vz799ygVsbW6WbzpoMUqBYm9q4Ri2 H9CM	21:32:45 0x058e20a0f 3044516...

The victim's 3-minute resend fired for the last two clips (21:31:15, 21:31:16) because no ack arrived in time. The guardian anchored using the victim's pre-signed anchor signature, so on-chain authorship stays with the victim's device.

Test 5: Victim reconnects PASS (after Bug 4 fix)

After airplane mode was turned off, the victim did not retry its uploads: the queued jobs were in exponential backoff, with next attempts 8–16 minutes away (Bug 4). After the fix the victim uploaded its own copies immediately on restart. They had the **same CIDs** as the guardian's uploads, which proves the guardian relayed the exact ciphertext:

Victim upload	CID	Same as guardian upload at
21:34:38	QmVaAUq...g6BW	21:29:52
21:35:59	QmWvKAX...MMWW	21:27:58
21:36:24	QmRQhfb...jREV	21:31:45
21:36:50	QmNxvKA...F2YW	21:32:25
21:37:07	QmVTAqF...H9CM	21:32:45

For none of these did the victim anchor again: its anchor job checked the chain first and found the guardian's anchor.

On-chain check (~21:37): EvidenceAnchored events read straight from the contract: **12 events, every CID exactly once.**

QmZbbdWU... (from an earlier run that day) · QmUpiXq5... · Qmc8LoAR... · QmRBjRvX... · QmWAvqqR... · QmWvKAXv... · QmVaAUqA... · QmRQhfbE... · QmNxvKAB... · QmVTAqFU... · QmUsqdcn... · QmXfTH9Q...

One more clip was anchored by the victim at 21:39:08 (tx 0xd1756b36...), after this check and shortly before the node was stopped, so it was not included in the duplicate check.

Test 6: Reverse relay (unplanned) PASS

The phones were paired both ways, so when the guardian's AI started a recording, the clip was relayed the other way. The guardian sent a key share and clip ac73608c... to the victim (21:36:27–28). The victim verified and stored it (21:36:30) and acked (21:36:32). The guardian anchored it itself (21:36:29, tx 0x49841e91...). The victim's later upload on its behalf (21:37:11, QmXfTH9Q...) correctly did not anchor again.

5. Bugs found and fixed (commit `d881ce1f`)

Bug 1: Camera unusable after the QR scanner

Symptom: every recording after a QR scan failed to start (`NullPointerException` in `ImageReader.getSurface()`). Earlier in the day, a recording that was running when the scanner opened could not be saved.

Cause: the scanner (`mobile_scanner`) takes the camera. Android closes the home screen's camera session, but the `CameraController` still reports itself as initialized.

Fix (`home_screen.dart`): release the camera before opening the scanner and reopen it afterwards. Block the scanner while recording ("Stop the recording first"). If a start fails on a stale controller, restart the camera and retry once. **Verified:** recordings started after scans, and the retry recovered automatically at 21:09:24 → 21:09:26.

Bug 2: Advertising silently stops after airplane mode

Symptom: an offline victim was invisible to its guardian.

Cause: advertising started while online (21:02:50). Turning on airplane mode turned Bluetooth off and Android stopped Nearby advertising without telling the app. The app's `_advertising` flag stayed `true`, so it never restarted.

Fix (`mesh_service.dart`, `outbox_service.dart`): `MeshService.refreshIfIdle()` restarts advertising and discovery while no peer is connected, at most once a minute (the first version restarted every ~10 s, which could cut off a connection being set up). It is called on every outbox run. **Verified:** the victim advertised offline from 21:14:17.

Bug 3: Guardians could never discover victims (present since Phase 4)

Symptom: with both phones paired, advertising and the guardian app open, no connection ever happened.

Cause: `startAdvertising` passed `serviceId: "com.justicechain.mesh"`, but `startDiscovery(_serviceId, strategy, ...)` passed the service ID as the *nickname* argument (the first parameter of `nearby_connections` 4.3.0), so discovery scanned the plugin default `com.pkmnapps.nearby_connections`. The two sides were on different services. Confirmed from Nearby's own logs on both phones.

Fix: pass the node ID as the nickname and `serviceId: _serviceId` by name. **Verified:** the guardian found the victim within ~1 minute of the fix (21:27:30).

Bug 4: Evidence waits out long backoff after reconnecting

Symptom: the victim was back online but did not upload.

Cause: jobs back off 30 s → 1 h. After 6 offline failures the next attempt was 16 minutes away, and reconnecting did not bring it forward.

Fix (`outbox_service.dart`): `OutboxService.retryAllNow()` makes every live job due immediately, on network reconnect and at startup. Permanently failed jobs are left alone. New unit test added. **Verified:** the victim uploaded all queued clips within ~1.5 minutes of restart.

Mitigation: AI 30-second cooldown (partial)

`ai_service.dart`: after a recording ends, the AI keeps listening but cannot auto-start another recording for 30 s. The manual SOS button is unaffected. This only reduces back-to-back false clips; it does not fix detection accuracy (see §7).

6. Environment and process issues

Issue	Impact	Resolution / lesson
<code>RPC_URL</code> is baked into the APK from <code>.env</code>	Every PC IP change needs a rebuild. Anchors failed with "Chain unreachable" when the APK had <code>10.159.143.50</code> but the PC was back on <code>10.35.251.51</code>	Rebuilt with the hotspot IP. A public testnet (Phase 7) removes this dependency
Hardhat node killed by Claude Code when the PC ran low on memory	Chain wiped; contract redeployed	Run the node in the user's own terminal. A fresh node plus one deploy always gives the same contract address
Large debug APK (216 MB) could not be shared over slow Wi-Fi	Delayed install on the guardian	Built per-ABI APKs (arm64: 105 MB). Installed over USB instead
Guardian data wiped during reinstall	Guardian lost its identity and pairing; re-registered and re-paired	The guardian had the split-ABI APK (versionCode 2001), so <code>flutter run</code> (versionCode 1) uninstalled it first. Lesson: use the same APK type on all test phones, or check version codes before installing
<code>flutter run --start-paused</code> left the app frozen when USB dropped	The app looked stuck; it was reinstalled	Stopped using <code>--start-paused</code>
Most app logs invisible in <code>flutter run</code> and <code>logcat</code>	Upload, anchor and mesh events could not be seen	Streamed <code>dart:developer</code> logs from the Dart VM service
Firewall rule for port 8545 needs admin rights	None in the end: Node.js was already allowed on the private network	Checked from the phone with <code>nc</code>

7. Open issues

AI false positives on normal speech

The 1D CNN model scored ordinary conversation as distress at 83–100%. Even with the 2-of-3-window rule it started recordings **at least 9 times** during the test (6 on the victim, 3 on the guardian, counted from the captured logs), often about 3 seconds after the microphone restarted. The trigger logic itself works (it resets after each recording); the model is the problem. **Next:** evaluate the model on speech or noise samples, add negative training data, consider raising the threshold or requiring 3 of 3 windows, and re-test.

Samsung: recording can fail to stop

At 21:38:24 on the guardian (Samsung A33), a recording that had started via the camera retry failed to stop (Failed to cleanly stop camera recording layers), and the clip was lost. The Samsung also needed the start-retry 3 times (21:25:15, 21:35:41, 21:37:51), so something keeps closing its camera. **Next:** find out what releases the camera on Samsung, and add a fallback that saves the partial file.

Minor

- **Duplicate guardian uploads:** a clip the victim resends after a missed ack is uploaded again by the guardian (same CID, no duplicate anchor). Harmless, but wastes bandwidth. The guardian could skip relays it already holds.
- **Simultaneous connect:** both phones run discovery, so they sometimes request each other at once ("Connection request failed", 21:35:54). One request succeeds, so it's harmless; it could be avoided by letting only one side connect.
- **Stale guardians:** the victim still trusts the guardian's old node ID (ac44225120047549), and relay jobs to it will never complete. There is no "remove guardian" option yet.
- **Not tested:** the two-guardian quorum ("Open & verify", "Send my piece") needs a third phone.

8. Commits and test status

Commit	Content	Tests
57d0f730	Shamir and outbox tests; guardian scanner context fix	62/62 pass
d881ce1f	Mesh service-ID fix, mesh refresh, outbox retryAllNow, camera reuse and retry, AI cooldown	63/63 pass; flutter analyze: 6 info lints (avoid_print in main.dart), no errors or warnings

Both commits are pushed to origin/local-blockchain--code (PR #3 against main). The local .env (not tracked) now points RPC_URL at http://10.35.251.51:8545 and CONTRACT_ADDRESS at 0x5FbDB...0aa3.

9. Suggested next steps

1. AI false positives: evaluate and retrain or tune the model (highest impact on usability).
2. Samsung camera stop failure: diagnose, and save partial recordings.
3. Three-phone quorum test: "Open & verify" and "Send my piece".
4. Delete local .enc files after upload and anchor are confirmed.
5. Phase 7: Polygon Amoy, which removes the PC/IP dependency for testers.
6. Phase 6: Android foreground service, so recording survives screen lock.
7. README and documentation; merge PR #3; build the release APK.